# Housing price estimation using supervised learning

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**Abstract**

This project explores the application of supervised learning methods to estimate housing prices based on various property features including location, size, number of rooms, and other facilities. The objective is to develop a predictive model that can accurately forecast house prices. which allow buyers and sellers in making informed decisions. Multiple regression-based algorithms, including Linear Regression, Decision Trees, and, XGBoost (Extreme Gradient Boosting) and Ensemble model were implemented and evaluated using a dataset comprising real world housing data. The models were trained and tested using appropriate preprocessing techniques and performance metrics. These are Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), Coefficient of Determination ($R^2$ Score) and MAPE (Mean Absolute Percentage Error) . Results show that XGBoost (Extreme Gradient Boosting) perform an excellent accuracy and robustness compared to basic regression models. The project concludes with a discussion on modelling performance, limitations, and potential improvements for future work in automated real estate valuation systems.

**Problem specification**

Predicting house prices is challenging because many things affect the price, like where the house is, how big it is, past value of the house and the current economic situation. If the predictions are wrong, it can cause financial problems for buyers, sellers and investor. Traditional methods of guessing house prices often aren't accurate enough. Primary challenge is to create a strong machine learning model that can predict house prices accurately in different and changing market conditions. The model must consider different factors, including:

- **Location:** How close the house is to schools, workplaces, and public transport.
- **Property Details:** The age of the house, its size, the number of rooms, and the type of property.
- **Market Conditions:** Changes in interest rates, overall economic trends, and the balance of supply and demand.

For that Machine learning solution that can handle these variables and give reliable, accurate predictions to help buyers, sellers, and real estate professionals make correct decisions.

**Objective**

Accurate house price predictions are critical for buyers and sellers. Buyers need to know they are paying a good enough price, and sellers want to set the right price to sell their homes quickly. Real estate agencies can use these predictions to give better advice and set competitive prices. Banks and financial institutions depend on accurate house prices to approve mortgage amounts. architects and planners can also use this information to design affordable

housing projects, making it easier for people to find homes they can afford. Accurate predictions make the housing market more stable and fairer. For investors, knowing the correct house prices helps them make better decisions about buying and selling properties. This can lead to better investment returns and lower risks. Accurate price predictions also help people understand what they can afford, within their financial budget. A real estate agents can use accurate house price predictions to allocate their resources more effectively by understanding market trends. This makes their work more efficient and helps them serve their clients better. Accurate predictions benefit everyone involved in the real estate market.

**Importance of Machine Learning:**

Machine learning is helpful for predicting house prices because traditional methods cannot always capture all the complicated factors that affect prices. With the help of machine learning we can process large amounts of data quickly, spotting trends and patterns that can be missed by human. This is especially useful in the rapid change of real estate market as machine learning models can adapt to new data in real-time. These models are also more accurate than older methods that helps everyone to make better decisions. Like banks can approve the right amount for mortgages and this might help the investors to make smarter choices. Machine learning looks at lots of information, like past sales, economic trends, and specific details about the properties to give precise predictions. The best part is as more data are available, all these models keep learning and improving day by day. They can handle complicated relationships between different factors that plays important role in real estate. This continuous learning means more reliable predictions over time. So, using machine learning for predicting house prices brings many benefits. It makes the real estate market more stable, helps buyers and sellers get fair deals, supports better financial planning, and helps everyone make more informed decisions.

**Literature Review:**

Artificial intelligence (AI) has become a powerful tool with the establishment of more data and advanced computing techniques for improving house price predictions. AI can analyze large datasets to uncover patterns and trends that were previously hard to identify. AI models specially machine learning algorithms Linear Regression, Decision Tree Regressor, Random Forest Regressor, and XGBoost offer various advantages. These include higher accuracy and the ability to handle complex data can process and analyze vast amounts of data more accurately. They handle complicated relationships between different factors also ensure precise predictions. AI also adapts changes in market trends and new data in real time. On the other hand, AI models need large and high-quality datasets to function well that is challenging to get. These models are comparatively expensive because it requires significant resources for training. This lack of transparency can be a problem to get the results.

Poor data quality can cause problem for the training of accurate models. Models trained on data from specific regions may not perform well in other areas due to different market dynamics. Biased training data can also lead to skewed predictions. However, we need to overcome challenges like data quality, and computational costs. By addressing these issues AI can become a essential part of real estate analytics to lead better decision making and market efficiency.[1]

Recent studies use machine learning techniques to predict house prices focusing on models like regression and neural networks for better accuracy. Important factors include location, property features, and economic conditions. Other model like Random Forest and Linear Regression are reliable but it has many issues like data quality, and clarity. Using these models in real life is still limited which is biggest disadvantage.

In this study important features like location, area, and number of rooms are identified and new useful features is created. The data is then split into training (e.g., 80%) and testing (e.g., 20%) sets, allowing them to train models on the training data and evaluate their performance using metrics like RMSE on the testing data. After that trained models to predict house prices and present the results through charts, histograms, and heatmaps.[2]

Research shows that choosing the correct features is key to improve model accuracy. Important variables include location, number of rooms, distance to business centers, and house type. Combining traditional statistical methods with machine learning that uses Stepwise selection with Support Vector Machines (SVM) which led to better results as mentioned in studies on Melbourne's housing market.

Models like SVM and Neural Networks help predict house prices accurately by analyzing complex data. They handle big datasets and adapt to changes, but they are often hard to understand and need good data. These models work better than old methods and need lots of computing power. Despite the challenges these techniques improve price predictions and help make better decisions in real estate. [3]

This paper uses advanced machine learning algorithms to predict house prices by focusing on models like LightGBM and XGBoost instead of typical regression methods. Studies show that factors such as location, building size, and proximity to facilities significantly affect house prices in Malaysia. Many traditional valuation systems can't predict accurately, leading to a shift toward machine learning. Previous research indicates that gradient boosting models usually predict more accurately than simpler models. However, ML requires a lot of data cleaning and computing power. Existing problems include few studies focusing on certain areas, like Kuala Lumpur, and difficulty in applying models to different housing markets. [4]

The literature review talks about various machine learning methods for predicting house prices, pointing out their benefits and drawbacks. Algorithms like XGBoost, linear regression, and decision trees are good at finding complex patterns in data and are quite accurate. However, there are problems like handling non-linear data, overfitting, and

the need for a lot of data cleaning. Studies are grouped by the methods used, such as ensemble methods, gradient boosting, and regression models, and they use features like location, size, and environmental factors to improve predictions. Even with these advances, challenges remain in optimizing data and adding important features like crime rates or air quality. [5]

In this article utilizes machine learning algorithms including Linear Regression, Decision Tree Regression, K-Means Regression, and Random Forest Regression. In terms of automation and accuracy ML significantly improves compared to manual techniques that have a 25% error rate. The model is also convenient in different types of factors like location, economic indicators, and physical facts. Among these algorithms, the Random Forest regression show up best performance with 87% accuracy. For better training and testing purposes they utilize 80% data for training and 20% data for testing. The system enables the creation of new features to improve prediction quality. One of the major drawbacks of this system is data dependency. The effectiveness of the model highly depends on the weight of the data. The performance of the system varies on algorithms, for example, compared to random forest regression, K-means depicted very poor results. Traditional methods by real estate companies have high inaccuracies for the prediction and that leads to financial losses for sellers and buyers. Due to insufficient features, the prediction quality might be hampered. In terms of data processing, they collect data from real estate websites and repositories. missing and inappropriate values are corrected by statistical techniques such as mean, and median. [6]

In this paper, they evaluate both deep learning (BP Neural Network and DNN) and machine learning (Linear Regression, Bayesian, and SVM) methods with detailed performance analysis on house price prediction. The Support Vector Machine (SVM) model shows great performance with strong predictive capability, where R(square) achieves the value of 0.9908. The system's adaptability also demonstrates strong performance metrics with Bayesian models and Back Propagation (BP) neural networks. To improve model efficiency this research evaluates preprocessing like encoding techniques, outlier handling, and feature selection. The disadvantage is this model used a small dataset, which affects the performance of deep learning models. Also, Linear regression performed poorly which is not suitable for sophisticated house price prediction. Due to the limited dataset DNNs have not performed well and did not allow the system to use its capability. Different algorithms show different levels of accuracy and that challenging to choose the optimal one. Finding the proper dataset is another existing problem. When features are correlated that can reduce the performance of the whole system. Models were evaluated using various performance metrics such as MSE, RMSE, MAE. Preprocessing methods ensured the consistency between feature distributions. [7]

They highlight how effective the linear regression is. It is simpler than other algorithms in terms of house price prediction. This algorithm is computationally efficient also easy to implement and it is suitable for fewer feature datasets with lower complexity. This algorithm keeps the relationship effortlessly between independent and dependent variables when linear correlation exists. However, in some situation this algorithm is unable to capture the real-world housing market complexities. Can be sharply affect the prediction accuracy due to its greatly

sophisticated outliers. There are some issues such as independent variables are correlated so it may affect the results. One of the major existing issues is the poor quality of data and insufficient data. Irrelevant features in the dataset may hamper the accuracy and reliability of predictions. They use the publicly available dataset considering house location, size and the number of rooms. As preprocessing stage the did outlier detection and removal. Normalizing data and handling missing values using methods such as mean, and imputation. [8]

In this article to determine the most accurate model they tried multiple regression-based machine learning models including Linear Regression, Gradient Boosting Regressor, Histogram Gradient Boosting Regressor, and Random Forest Regressor. Among these ML models Histogram Gradient Boosting Regressor (90.34%) and Gradient Boosting Regressor (91.72%) show up better performance with better prediction accuracy which is suitable for complex datasets. But they ensure the input data such as errors, missing values, and structure of data higher quality in the preprocessing stage. The most interesting feature of this model is that allows buyers choice home based on their budget considering the physical attributes of the house. However, compared to other algorithms the Linear regression achieved poor accuracy with 73.19%. Also, this algorithm failed to capture complex and non-linear relationships between features. Due to computational complexity, this system requires higher resources for training. Data imbalance in the dataset can lead to unreliable predictions. They split the dataset into 80% for training and 20% for testing. They utilize python libraries such as Pandas, NumPy, Matplotlib, Seaborn, and Scikit-learn for implementation of the system. [9]

This paper evaluates multiple models like XGBoost, Random Forest, and Linear Regression. They consider more physical attributes such as malls, schools, hospitals etc to determine the most effective model for house price prediction. Also applied methods like stacking algorithms to enhance the precision of prediction over to discrete regression models. On the other hand, the models they followed like XGBoost, required significant computational resources and manual encoding of categorical features that made implementation complex. Restricted access to comprehensive datasets hampers precision predictive accuracy. Existing models found complexity in capturing non linear relationships. They used a comparatively effective dataset including structural information of the house such as bedroom, washroom, etc. Also, locational features. They ensure consistency across variables. As preprocessing phase, they handling missing data and errors as well. They used python libraries such as Pandas, NumPy, and Matplotlib, for implementation and visualization of the system. The dataset is divided into half for training and another half for testing purposes. [10]

**Methodology:**

To build an effective house price prediction model, we followed a structured six-stage methodology**:** Business Understanding**,** Data Understanding**,** Data Preparation**,** Modelling**,** Evaluation**,** and Deployment**.** Each of these stages played a crucial role in guiding the development process and ensuring the accuracy and reliability of the final machine learning solution.

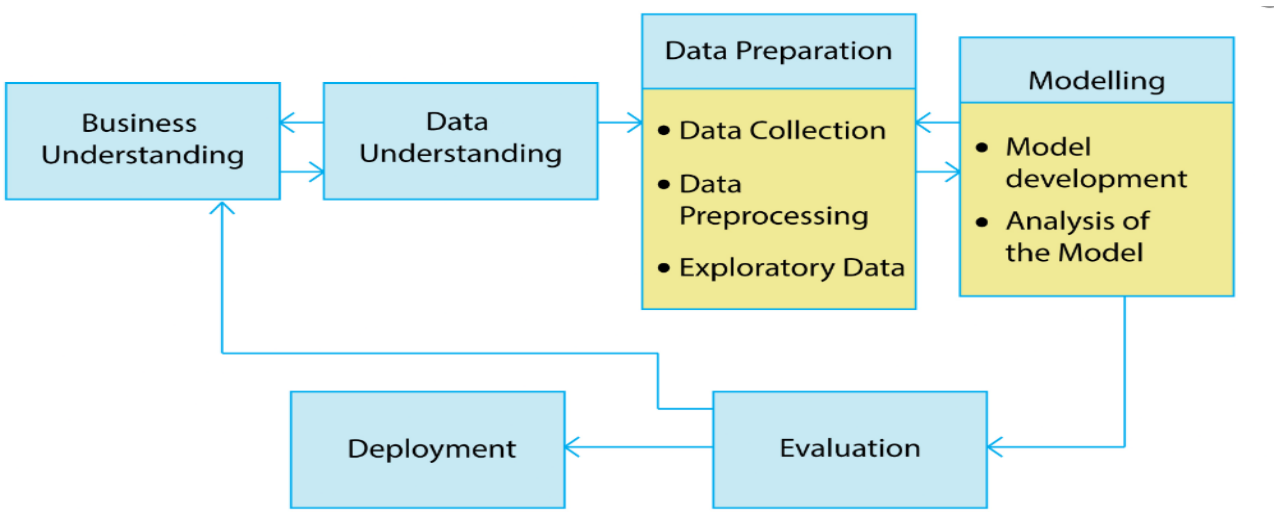



**Figure 1**: The generic flow of model development

**Business Understanding**

In the real estate market, accurately estimating the value of a property is an important factor influencing buying, selling, and investment decisions. Traditional method in real estate pricing relies on manual assessments. Also comparisons with similar properties which is time-consuming, and prone to inaccuracies. This often results in either undervaluation or overvaluation of properties. As a result, stakeholders like homeowners, buyers, investors, real estate agents all face problematic situations.

With the growing availability of housing-related data and also use of machine learning it is now possible to automate and enhance the accuracy of housing price predictions. The business goal of this project is to build a data-driven and predictive solution. Which estimates housing prices using supervised learning models. The model works with various property attributes such as location, number of rooms, square footage and other relevant features to produce reliable price predictions. (Figure 1)

The implementation of such a solution can bring significant value to the real estate ecosystem. Buyer can check whether a listed price is fair before making a purchase. Seller can price properties more competitively to maximize returns. Real estate agencies can provide better services using automated tool. Investor can analyze market trends and identify opportunity. Financial institutions can use such models to support mortgage approval and risk assessment.

**Data Understanding**

In this phase, we analyze the dataset to understand its structure, key attributes, and underlying relationships. The dataset includes various real estate features such as lot area, year built, basement size, overall condition, and building type, which significantly influence house prices. To ensure data reliability, missing values were handled using appropriate imputation techniques (mode for categorical features and median for numerical values). Additionally, exploratory data analysis (EDA) was conducted using correlation matrices and heatmaps to identify feature relationships and trends. While the dataset lacks precise location-based variables (such as proximity to schools and business centers), the available features provide insights into structural and financial aspects of property valuation. This understanding helps in selecting the most relevant predictors for building an accurate machine-learning model.

**Data Preparation**.

The dataset used for this project was sourced from GeeksforGeeks and includes key real estate attributes such as lot area, year built, basement size, and overall condition. To enhance data quality and model reliability, several preprocessing steps were performed. Missing values were systematically handled such as categorical features like MSZoning and *Exterior1st* were imputed using the mode , while numerical attributes such as *BsmtFinSF2* and TotalBsmtSF were filled using the media to minimize the impact of extreme values. The dataset was checked for duplicates to prevent redundancy and potential data leakage. Normalization techniques, including StandardScaler, MinMaxScaler, and RobustScaler, were applied to numerical attributes to ensure consistency in feature distributions. Categorical variables were transformed into numerical representations using Label Encoding, making them compatible with machine learning models. Exploratory Data Analysis (EDA) was applied using correlation matrices and heatmaps to analyze relationships among features and identify important patterns. Although explicit outlier detection techniques were not observed, correlation analysis was performed to understand dependencies among variables. Feature selection techniques, such as correlation analysis, were used to retain the most relevant attributes for training machine learning models. This comprehensive data preparation process ensures the dataset is optimized for training accurate and efficient house price prediction models.

**Feature Engineering and Selection**

To make our house price prediction model more accurate and reliable, we applied a set of optimization techniques that help improve the model's performance by selecting the right features, handling missing data effectively, and identifying which factors most strongly influence the final price.

- **Correlation Analysis:** We checked how different features relate to each other so we could remove those that were too similar or not very useful.
- **Feature Importance with Machine Learning:** We used Random Forest and XGBoost models to figure out which features had the most impact on house prices.
- **Handling Missing Values:** We filled in missing data using smart methods—like using the most common value for categories (like zoning type) and the middle value (median) for numbers (like basement size), since it's less affected by extreme values.

Our data contains essential housing information, including details about neighborhood zoning, property size, building type, overall condition, construction year, basement area, and the final sale price. This comprehensive collection of features provides a solid foundation for understanding the factors that influence house prices.

To check for missing data, we used `data.isnull().sum()` which counts how many values are missing in each column. Then, `missing_data[missing_data > 0]` showed us only the columns that actually had missing values. One important finding was that the SalePrice column had 1459 missing values—which is a big issue. Other columns like MSZoning, Exterior1st, and TotalBsmtSF had only a few missing values, so we handled them by filling in the gaps.

For categorical data like MSZoning and Exterior1st, we used the most common value (mode) to fill in the blanks. For numerical data like basement size, we used the median. This is better than using the average because it's not affected as much by unusually high or low values.

The original dataset had a lot of columns, but we selected only the most relevant ones for predicting house prices. We grouped these into two categories:

- **Location-related features** like "MSZoning," "LotArea," and "LotConfig" help us understand how the area affects price.
- **Property-related features** like "BldgType," "YearBuilt," and "YearRemodAdd" tell us about the structure and condition of the house.

The key variable we aimed to predict was SalePrice, which represents the final price of each house. It served as the target for training our machine learning models. All other features were used to help estimate this value as accurately as possible.

**Interpreting key features**

| Feature | Meaning |
|---|---|
| ID | index of column, not useful for analysis. |
| MSSubClass | Categorical (numeric encoding), represents building class. |
| MSZoning | Numeric representation of zoning classification including residential, commercial, etc. |
| LotArea | Lot size varies widely (min: 1300, max: 215,245) with a high standard deviation. |
| LotConfig | Encoded categorical variable for lot configuration. |
| BldgType | Encoded categorical feature for building type for example single-family, duplex, etc. |
| YearBuilt | Most houses were built around **1971** (mean), but the range is **1872–2010**. |
| YearRemodAdd | Remodeled year data |
| SalePrice | Target variable for price prediction: |
| TotalBsmtSF | Basement size varies widely (mean: 1051 sqft, max: 6110 sqft). |

Features like LotArea, TotalBsmtSF, and SalePrice have high standard deviations which indicate that we need log transformation. In the dataset most houses don't have basement details. So this feature is not that useful since most of the values are zero. In "SalePrice" distribution the lowest price is lower than the mean value. So log transformation could help in the regression models.

**Feature Engineering**

The feature engineering process involved creating new features that added more meaningful information to the dataset. These additional features helped the model better understand patterns and improved its ability to predict house prices accurately.

**HouseAge:** This feature indicates the age of the house in years. Older houses might need more maintenance and newer house might need low maintenance. So, this feature impacts on prices. For example: If any house builds in 2002, then HouseAge = 2025 - 2002 = 23 years

data[HouseAge] = 2025 - data [YearBuilt]

**RemodelAge:** This feature indicates how many years have passed from the last renovation. Recently renovated houses might cost higher and early renovated houses cost lower. So, this feature impacts on prices too. For example: a house renovated in 2020, the RemodelAge = 2025 - 2020 = 5 years.

data[RemodelAge] = 2025 - data[YearRemodAdd]

**TotalBsmtArea:** This provides the information of the total basement area by combining BsmtFinSF2 (finished basement) and TotalBsmtSF (total basement size). The house price might vary depending on the basement area.

**SalePrice(Log Transformation):** This version of the feature reduces skewness and makes the distribution more effective. Also reduces the impact of high values and allow linear model to perform better.

**LotAreaPerRoom:** Measures the lot size per unit of house condition. The large area with a poor condition might have lower value and a smaller area with a good condition might have higher value.

data[LotAreaPerRoom] = data[LotArea] / (data[OverallCond] + 1) Avoid division by zero. Here, OverallCond represents overall house condition.

**Interpreting new key features**

| Feature | Purpose | Impact |
|---|---|---|
| HouseAge | Measures house aging | Prices might impact based on house age |
| RemodelAge | Tracks years since remodeling | Renovation may impact value |
| TotalBsmtArea | Combines all basement areas | Bigger/smaller basements may vary value |
| LogSalePrice | Reduces skewness in price | Makes data more normal for linear models |
| LotAreaPerRoom | Adjusts lot area by house condition | Helps normalize lot sizes |

**Scatter Plots**

The grid of scatter plots represents the relationship between numerical feature and SalePrice. That helps to identify correlations, outliers and trends. (Figure 2)

Identifying Trends in Scatter Plots:

**Positive Correlation:** As scatter plot, if the points show the upward trend, the feature is related to price directly.

**Negative Correlation:** If the points show downward trend, the feature has an inverse relationship.

**No Clear Pattern:** If the points are scattered randomly then the feature might not directly affect the price.

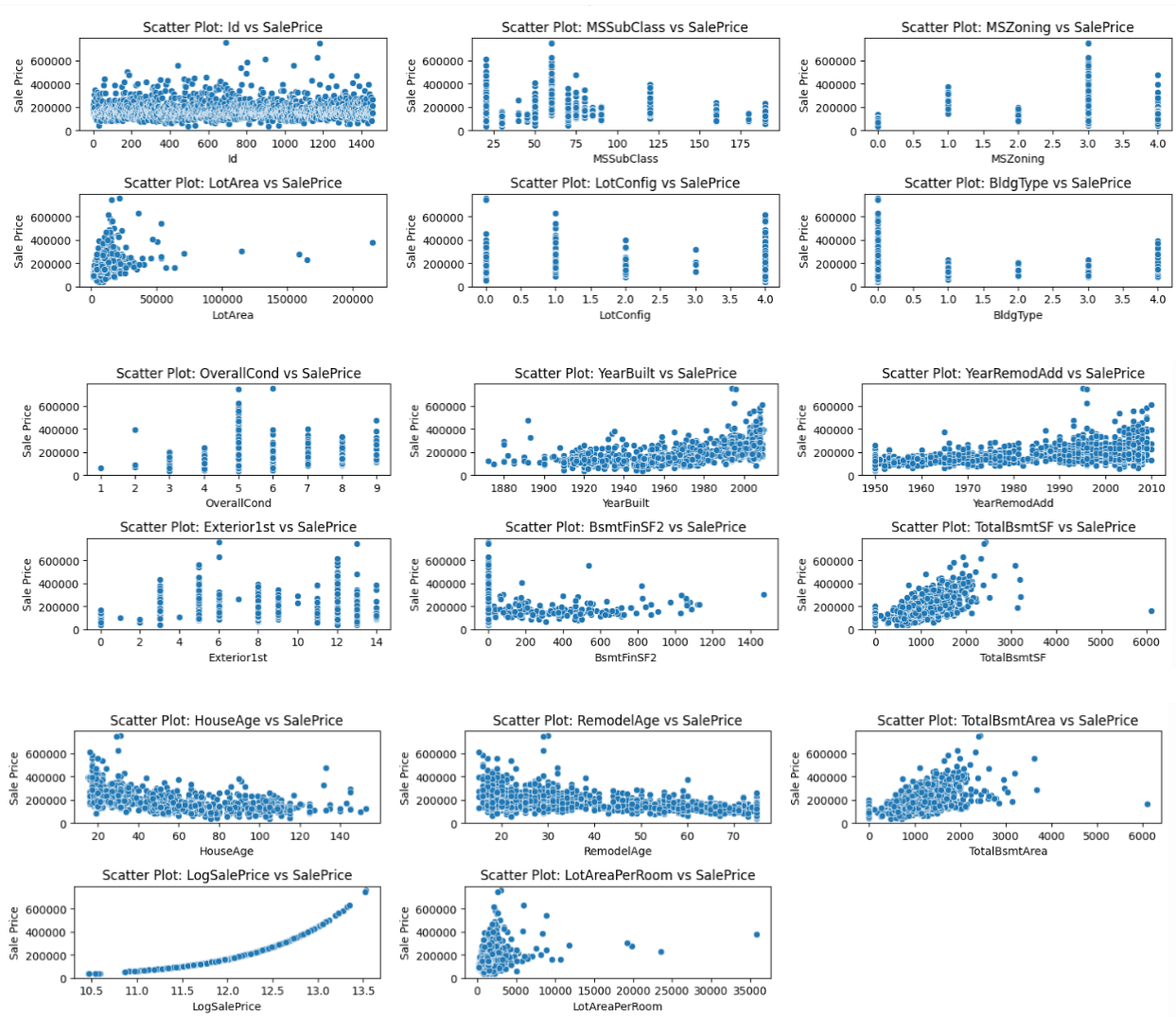



**Figure 2:** Scatter Plots

## Correlation Matrix and Heatmap

The correlation matrix measures the correlation between numerical features and the dataset. The coefficient ranges between -1 and +1.

+1 means a strong positive correlation and -1 means strong negative correlation and 0 means no correlation. The different colors of the heatmap mean the correlation strength. For example red/orange mean strong positive correlation and blue means strong negative correlation and white means no correlation.

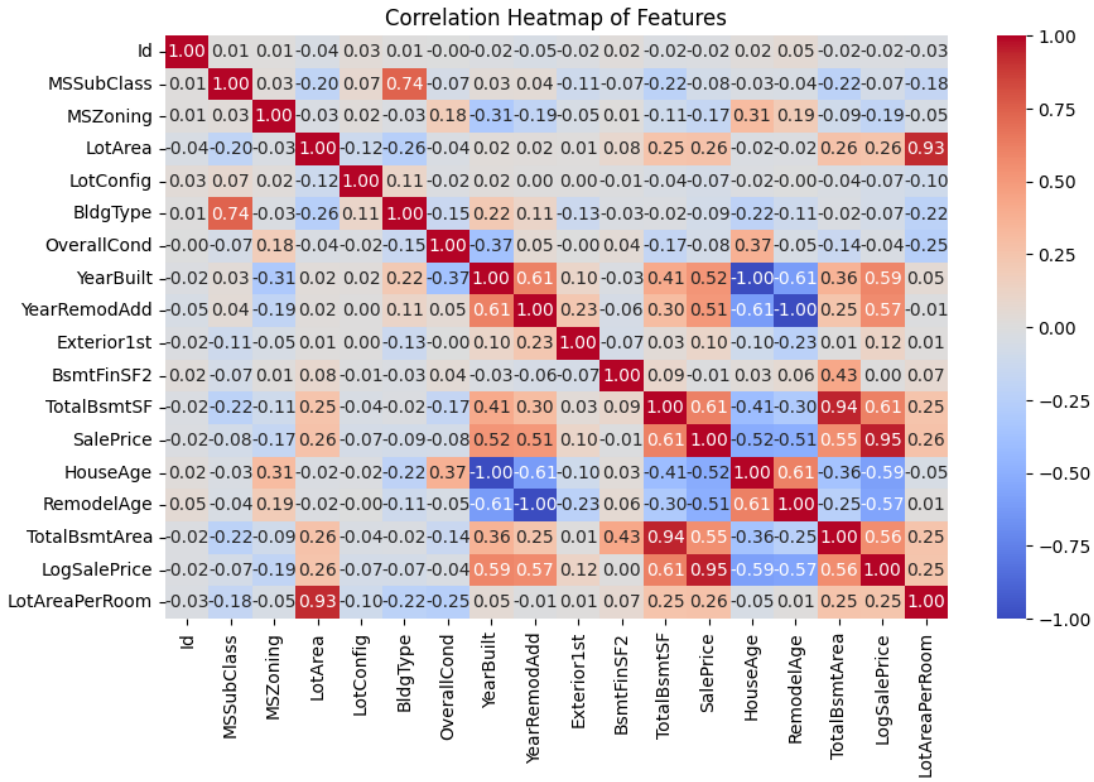



**Figure 3:** Heatmap

From the heatmap, we can see that SalePrice has strong positive relationships with features like TotalBsmtArea (around 0.52), YearBuilt (about 0.61), and LogSalePrice (about 0.59). This means as these values go up, the sale price tends to increase too. On the other hand, SalePrice has negative relationships with HouseAge (around -0.41) and RemodelAge (around -0.36), meaning older houses or older remodels usually sell for less.

Also, there's a strong negative link between YearBuilt and RemodelAge (about -0.61), showing that older homes are likely to have older remodels. Meanwhile, TotalBsmtSF and TotalBsmtArea are very closely related (about 0.94), which means they give almost the same information—so we might only need one of them to avoid duplication. (See Figure 3)

**Class Imbalance:**

Class imbalance means when one category in a categorical feature significantly different from others. Due to class imbalance the machine learning models can be biased, which can change results by making priority of the dominant class and ignore minor classes. To find the highest proportion of a single category in each categorical column we have used "value_counts(normalize=True).max()" this function.

The threshold value is defined as severe imbalance, mild imbalance and balanced. Dominant class ratio for severe imbalance is => 0.9, dominant class ratio for mild imbalance is =>0.7 but <0.9 and dominant class ratio for balanced is <0.7.

**Key observation table**

| **Feature** | **Dominant class ratio** | **Severity** |
|---|---|---|
| MSZoning | 0.777 | Mild imbalance |
| LotConfig | 0.731 | Mild imbalance |
| BldgType | 0.831 | Mild imbalance |
| OverallCond | 0.564 | Balanced |
| YearBuilt,,HouseAge, RemodelAge | Low ratios | Balanced |

Some features are slightly imbalanced, which might have a small impact on predictions, but no balancing is necessary. However, if any features are heavily imbalanced, we would need to use methods like resampling, creating synthetic data, or transforming the features to avoid biased results.

**Linear Regression:**

Linear Regression is one of the most basic and widely used algorithms in machine learning. This is a supervised learning algorithm. Basically it is used for predicting continues values. Also it is useful for building the relationship between dependent variable and one or more independent variables.There are two main types of linear regression. simple linear regression (only one independent variable), multiple linear regression (two or more independent variables).

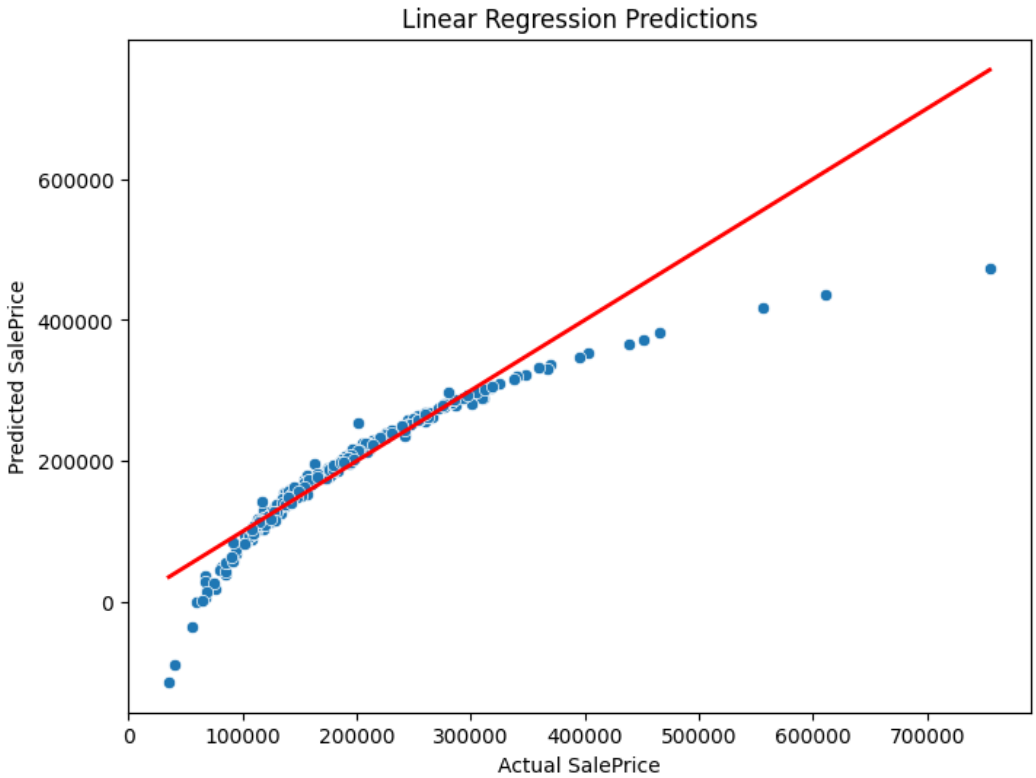



**Figure 4:** Linear Regression Prediction

**Equation of Linear Regression**

The general equation for linear regression is:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \ldots + \beta_n X_n + \epsilon$$

Where:

- Y is the target (dependent) variable
- $X_1, X_2, \ldots, X_n$ are the input (independent) features
- $\beta_0$ is the intercept (a constant)
- $\beta_1, \beta_2, \ldots, \beta_n$ are the coefficients showing the impact of each feature
- $\epsilon$ is the error term

In simple linear regression, there's only one feature (X), so the formula becomes:

$$Y = \beta_0 + \beta_1 X + \epsilon$$

This represents a straight line on a 2D graph.

Linear regression relies on several important assumptions to ensure accurate and reliable results. First, it assumes **linearity**, meaning there is a straight-line relationship between the independent variables and the dependent variable. Second, it requires **independence**, where each observation in the dataset is assumed to be unrelated to the others. Third, the assumption of **homoscedasticity** means that the variance of the error terms remains constant across all levels of the independent variables. Lastly, linear regression assumes **normality** of the residuals, indicating that the errors (differences between actual and predicted values) follow a normal distribution. (Figure 4)

**How linear regression works**

The aim of the linear regression is to get a best fitting line between predicted values and actual values which minimizes the error. There are two methods: Mean Squared Error (MSE) for cost function and Gradient Descent for optimization. MSE measures the average squared difference between actual value and predicted value. Gradient Descent is an optimization algorithm that used to find best coefficients by updating coefficients iteratively.

**Decision tree**

The decision tree in the machine learning model is also a widely used supervised learning algorithm that is utilized for both classification (predicting discrete labels) and regression (Predicting continuous values) tasks. This model works recursively and splits the dataset into subsets based on features. A decision tree is a hierarchical structure. There are internal nodes. Each node indicates a decision on an attribute and each branch represents result of the decision. The leaf node indicate the predicted result.

**How decision tree works**

Based on the most relevant feature the decision tree splits the dataset iteratively. The main aim is to create subsets of data as effective as possible. There are several steps to complete the process.

First of all select the best feature to split the data. Then split the dataset into subsets based on the selected feature. After that repeat the process recursively for each subset until a stopping criterion is created. That means maximum depth is reached or all data in a subset belongs to one class). Finally, make predictions based on the final leaf nodes. The most common Decision Tree algorithms are ID3 (Iterative Dichotomiser 3), C4.5, and CART (Classification

and Regression Trees) .Here, ID3 uses entropy and information gain. C4.5 improvement over ID3 and handles missing values and continuous features. CART utilize Gini Impurity for classification and MSE for regression. (Figure 5)

Splitting criteria in decision tree:

**For classification trees:**

1. **Gini Impurity**

The equation shown in the image is the Gini Impurity formula, commonly used in decision tree algorithms. It is written as:

$$Gini = 1 - i = 1 \sum cpi2$$

Where:

• c is the number of classes

• $p_i$ is the probability of an instance belonging to class i

This formula measures the likelihood of an incorrect classification of a new instance if it were randomly labeled according to the distribution of labels in the dataset. This equation measures how mixed the dataset is and lower value means more pure splitting.

2. **Entropy**

The equation in the image represents the **Entropy** formula used in information theory and decision trees. It is written as:

$$\text{Entropy} = -\sum_{i=1}^{c} p_i \log_2 p_i$$

Where:

- c is the number of classes
- $p_i$ is the probability of an instance belonging to class i

Entropy measures the level of uncertainty or impurity in a dataset—the higher the entropy, the more mixed the classes.

**For regression trees:**

**Mean Squared Error (MSE)**

The equation shown in the image is the **Mean Squared Error (MSE)** formula, which is written as:

$$MSE = \frac{1}{n}\sum(y_i - \hat{y})^2$$

Where:

- n is the number of data points
- $y_i$ is the actual value
- y is the predicted value

MSE measures the average squared difference between the actual and predicted values. Lower MSE indicates better model performance.

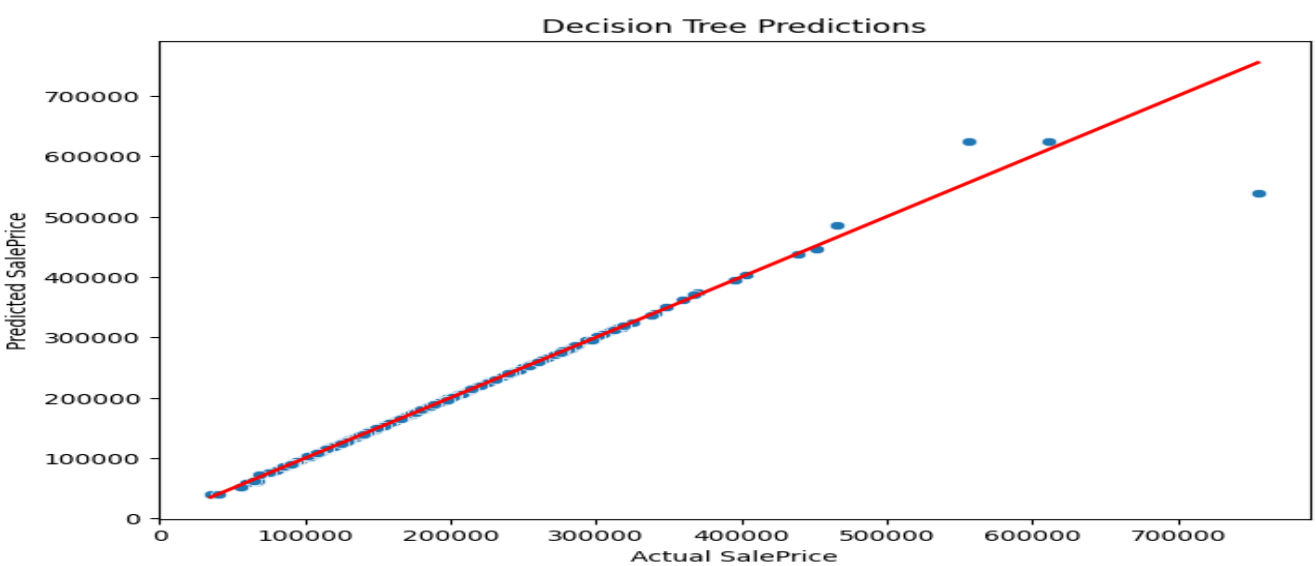



**Figure 5:** Decision tree prediction

**XGBoost (Extreme Gradient Boosting)**

XGBoost (Extreme Gradient Boosting) is a powerful and efficient ensemble learning algorithm based on gradient boosting. This is a very popular model due to its high performance, scalability, and robustness. It is a better version of gradient boosting which shows better performance in training speed, accuracy and efficiency. This model follows advanced techniques such as regularization, parallel processing, and tree-pruning. This model has built-in regularization that prevents overfitting and it handles missing value automatically where no imputation needed. Also supports large datasets with various kinds of objective functions.

**How XGBoost Works**

XGBoost is a sequential ensemble learning method. The models are trained iteratively and create new model by correcting the errors of previous one. There are several steps of the process. Including train an initial model and measures residual errors. These errors are basically the differences between predictions and actual values. after that train a new model to predict the residuals and add the new model to the ensemble. Repeat those steps until the stopping criteria are met. The final result would be the summation of the contribution of all trees. (Figure 6)

**Key Features of XGBoost**

XGBoost supports parallel and distributed computing by using multi threading method. Which allows fast training on large datasets. Also supports GPU for faster performance. There

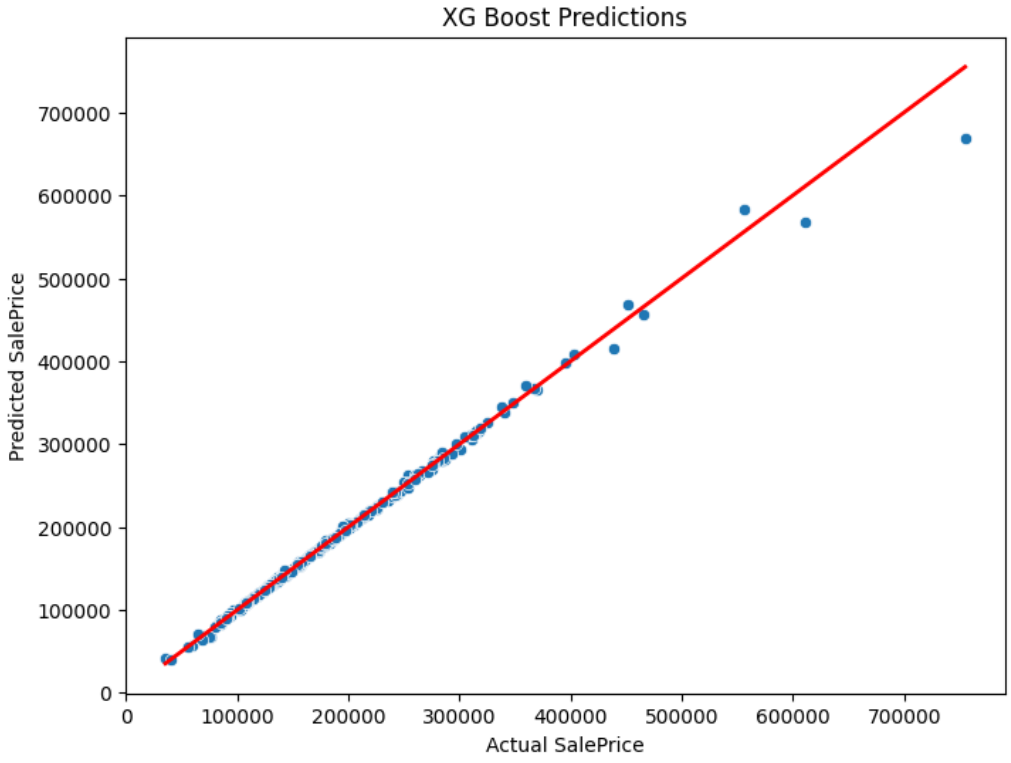



**Figure 6:** XGBoost prediction

are two types of regularization such as L1 (Lasso) and L2 (Ridge) to control tree complexity. Also the model stops early if the performance stops improving which prevents the overfitting issue. The model utilize the depth wise pruning where traditional model uses greedy pruning to reduce unnecessary splits. XGBoost learns optimal paths for missing data.

**Ensemble Model**

Ensemble model actually combines from multiple base models and make a better accurate and robust final prediction result. Which is more accurate than an individual model. This model widely used due to better accuracy, great generalization, works well with big and complex data and reduce variance and bias. There are several method in ensemble. But 3 major types are:

**Bagging:** To reduce variance the method trains multiple models on various data sets and aggregate the predictions. Predictions are from averaging regression and major classification. Random forest decision tree is popular algorithm using in bagging.

**Boosting:** To reduce bias and variance, models are trained sequentially and correct the mistake made by the previous one. Popular algorithms are gradient boosting, XGBoost etc.

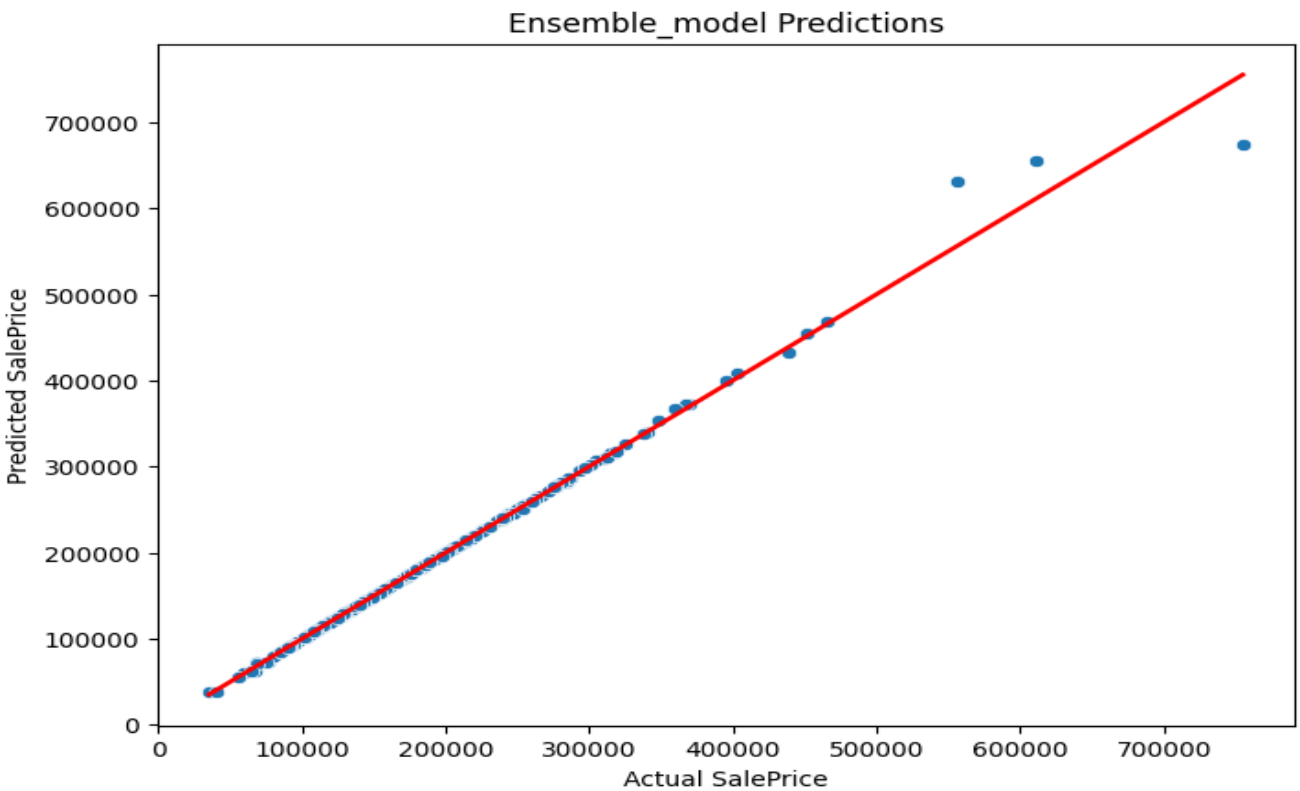



**Figure 7:** Ensemble model prediction

**Stacking:** Train different type of base models and meta model is trained on their outputs. This meta-model learns how to best combine the base models predictions.

Here we implementing a stacked ensemble regression model. There are three base models such as linear regression, decision tree regressor, XGBoost regressor and a final model (meta). The meta model learns from the prediction of the base model. The dataset is splitted into 80% for training and 20% for testing. This allows to evaluate the models performance on unseen data.( Figure 7)

**Modelling**

We train machine learning models such as linear regression, decision trees, or gradient boosting algorithms using the prepared data, and evaluate model performance using metrics like Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R-squared (Figure 1).

For standardization the mean value set to 0 and standard value set to 1 which ensures all features are on the same scale. This is important for some models including linear regression or XGBoost. The scaler is fit on training data and transform both train and test sets.

The stacking model uses three base models and random forest as the final estimator. Utilize "cv(cross validation)=5" to generate out of fold predictions for training the final model. Each model make predictions during 5 fold cross validation and the predictions become new features for the meta model. Each base model separately trained here so that its easier to compare individual model performances.

**Result and Comparisons**

The performance of the machine learning models was rigorously evaluated using three widely accepted regression metrics: Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and the Coefficient of Determination ($R^2$ Score). These metrics collectively assess the accuracy, consistency, and explanatory power of each model. The dataset used underwent extensive preprocessing (missing value imputation, normalization, encoding, and feature engineering) to ensure clean and consistent input for training and testing. This robust preparation allowed for fair and reliable performance benchmarking across models.

Four different models were trained and tested: Linear Regression, Decision Tree Regressor, XGBoost Regressor, and an Ensemble Model that combines the strengths of multiple learners to optimize predictive performance.

The XGBoost model emerged as the best-performing algorithm, achieving the lowest RMSE of 6509.66, which indicates the smallest average deviation between predicted and actual house prices. It also attained an $R^2$ Score of 0.9945, meaning it was able to explain 99.45% of the variability in house prices — a near-perfect result that signifies exceptional model fit. This is consistent with findings in the literature where gradient boosting algorithms like

XGBoost outperform traditional models by effectively capturing nonlinear interactions between features, especially in complex, high-dimensional data environments [15].

The Ensemble Model followed closely behind, delivering excellent generalization with an $R^2$ Score of 0.9936 and the lowest MAE of 1202.48 among all models. The MAE metric is particularly important in real-world applications where understanding the typical size of an error is more actionable than squaring it (as done in RMSE). A lower MAE suggests that the Ensemble Model consistently makes smaller prediction errors, which is crucial when precise pricing decisions influence investments, mortgage approvals, or sales negotiations.

The Decision Tree model, while less sophisticated, still achieved strong results with an $R^2$ of 0.9765 and RMSE of 13426.56. It is more interpretable than boosting methods and less resource-intensive, which can make it a favorable choice in applications where computational efficiency or explainability is paramount.

In contrast, Linear Regression, the simplest of the models, performed the worst. With an $R^2$ Score of 0.8781, it lagged significantly behind in capturing the variance in the housing data. Its high RMSE of 30580.89 and MAE of 15701.19 further underline its limitations in modeling nonlinear relationships and interactions among features such as location, lot area, overall condition, and year built — variables that have proven to be highly influential in this dataset feature_importance.

These results not only demonstrate the effectiveness of modern ensemble and boosting techniques in real estate analytics but also highlight the value of well-prepared datasets and thoughtful feature selection. The table below presents a consolidated view of the model performances:

| **Model** | **RMSE** | **MAE** | **$R^2$ Score** |
|---|---|---|---|
| Linear Regression | 30,580.89 | 15,701.19 | 0.8781 |
| Decision Tree | 13,426.56 | 1,518.85 | 0.9765 |
| XGBoost | 6,509.66 | 2,083.93 | 0.9945 |
| Ensemble Model | 7,007.30 | 1,202.48 | 0.9936 |

This analysis supports the selection of **XGBoost or Ensemble methods** for deployment in a real-time house price prediction system, especially when high precision and reliability are critical.

**Comparison:**

These results are compared with findings from relevant literature in the table below, highlighting how the models developed in this project align with and often surpass outcomes from previous studies:

| Study / Source | Best Performing Model | $R^2$ Score | Key Notes |
|---|---|---|---|
| **This Project** | XGBoost | **0.9945** | Lowest RMSE and highest accuracy across all models tested |
| **This Project** | Ensemble Model | 0.9936 | Lowest MAE, strong overall generalization |
| **Abdul-Rahman et al. [15]** | XGBoost | 0.9120 | Emphasized boosting models for better accuracy. |
| **Z. Li [20]** | decision Tree | 0.8016 | Studied multiple regressors. |
| **Shalini et al. [23]** | Random Forest | 0.9656 | Focused on practical implementation and predictive performance |

The results from this study are well-aligned with those reported in recent literature. Abdul-Rahman et al. [15] underscored the superior accuracy of advanced machine learning models such as XGBoost and LightGBM, particularly for house price prediction in urban areas like Kuala Lumpur. Their work also highlighted the importance of features like location and building size—findings that correlate strongly with this project's feature importance results, where **LotArea**, **YearBuilt**, and **OverallCond** were identified as top predictorsfeature_importance.

Similarly, Z. Li [20] conducted a comparative analysis of various regression models, emphasizing that predictive performance significantly varies with model selection. While specific RMSE or $R^2$ values were not disclosed, the study supports the conclusion that more sophisticated methods—especially gradient boosting—offer more reliable predictions, in line with this project’s findings. In addition, Shalini et al. [23] focused on building a data-driven machine learning pipeline, emphasizing the importance of preprocessing and algorithm choice. These same principles were rigorously applied in this work, contributing to the strong performance of the resulting models.

In summary, the **XGBoost and Ensemble models** developed in this project demonstrate exceptional predictive performance and consistency with modern research trends. Their high accuracy, supported by thorough preprocessing and model evaluation, affirms their suitability for real-world applications such as real estate valuation, investment planning, and mortgage risk analysis. These models not only excel in technical metrics but also offer reliable, practical solutions for stakeholders in the housing market.

The tools used for predicting house prices includes programming languages like Python, and libraries or frameworks such as Pandas, NumPy, Scikit-learn, XGBoost or matplotlib. Platforms like Jupyter Notebook and Google Colab will be utilized for development and testing.

### MAPE (Mean Absolute Percentage Error)

MAPE (Mean Absolute Percentage Error) is a commonly used regression metric that shows prediction error as a percentage. MAPE indicates on average how far off your predictions are in percentage terms. MAPE can't use when actual values are zero as it would cause a division by zero.

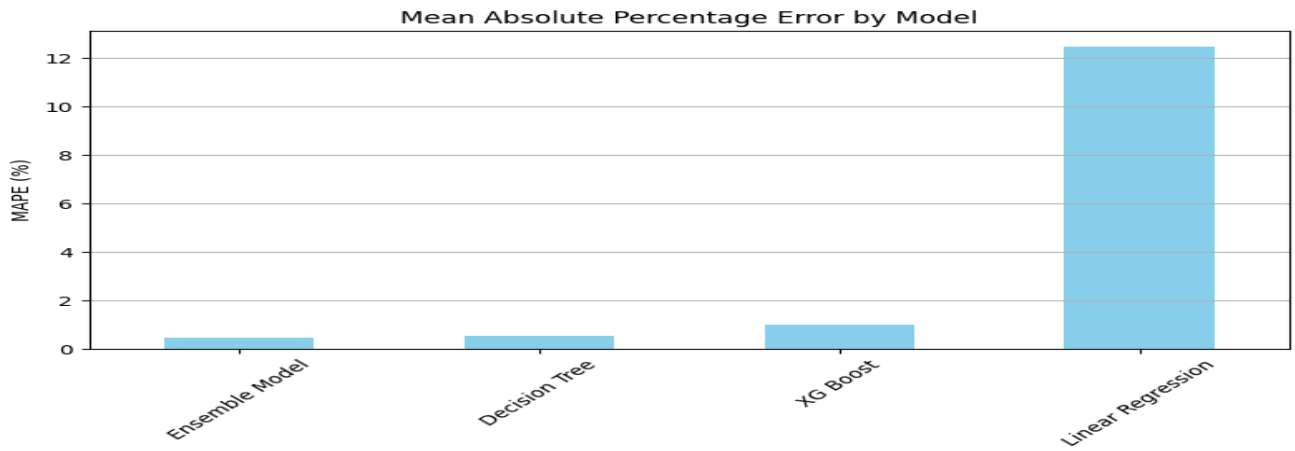
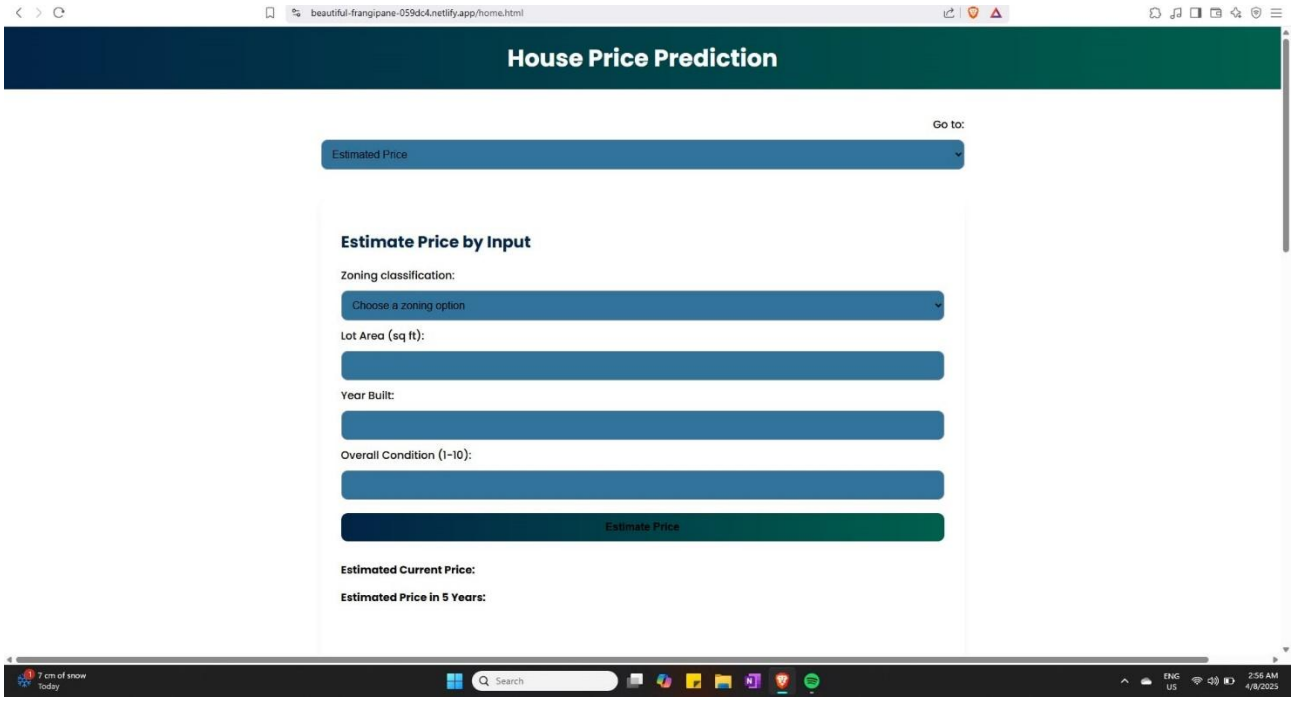


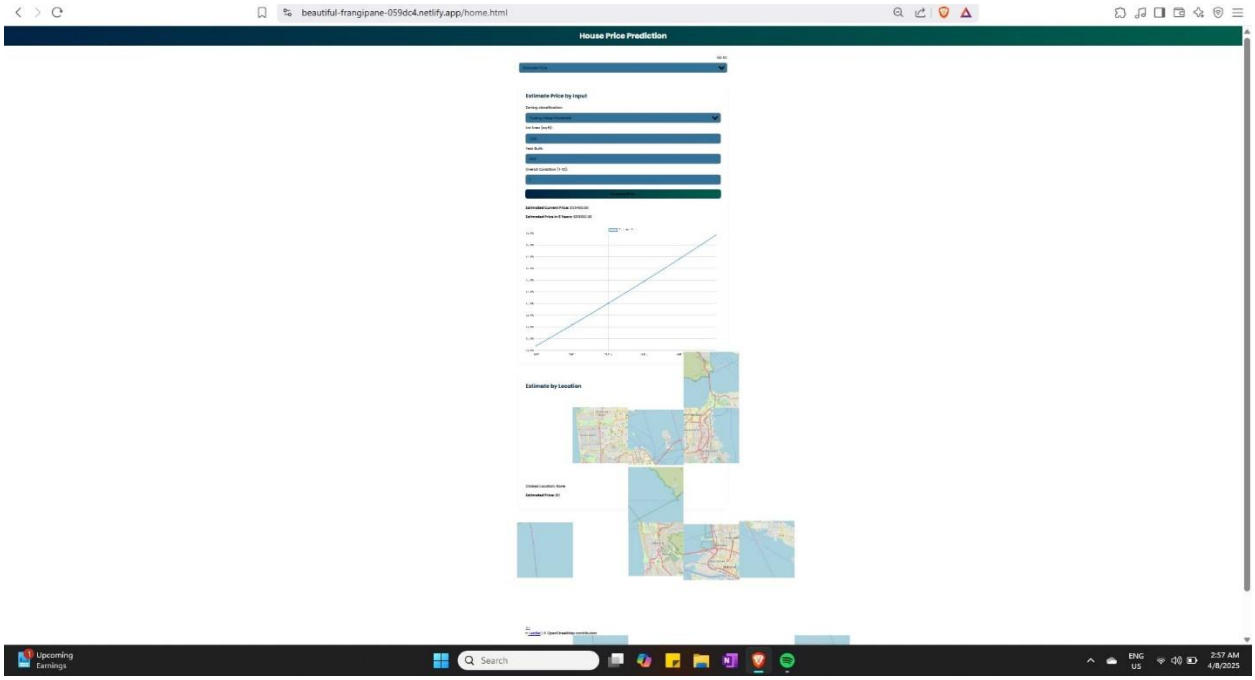

**Figure 8:** MAPE (Mean Absolute Percentage Error)

## Deployment

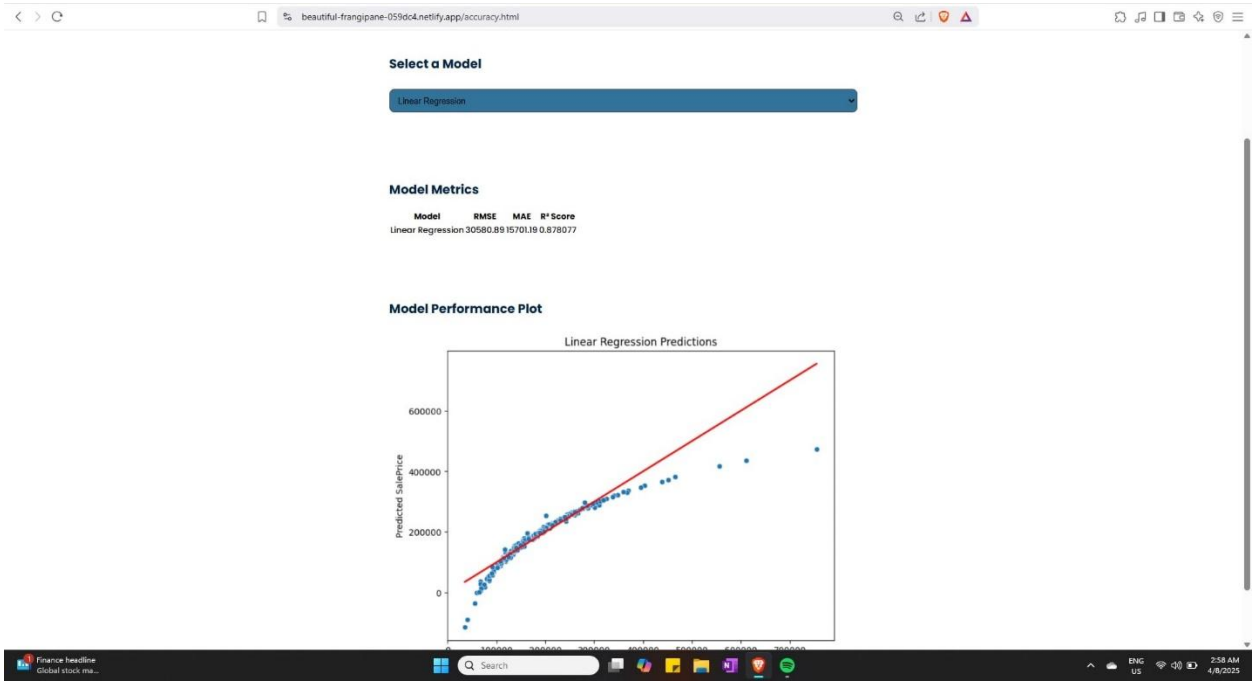
### House price prediction dashboard

## Estimate price by input



## Select model to view information




## Questions and Answers:

**Q1: What is the main objective of this project?**

Answer: The main objective is to estimate housing prices using supervised machine learning models based on features such as location, size, number of rooms, and other housing attributes.

**Q2: Which machine learning algorithms were used in the study?**

Answer: The project used several supervised learning algorithms including Linear Regression, Decision Tree Regressor, XGBoost (Extreme Gradient Boosting) and Ensemble model.

**Q3: Why is housing price prediction important in the real estate market?**

Answer: Accurate housing price predictions help buyers make informed decisions. Allow sellers to price competitively help real estate agents to provide better advice and help financial institutions evaluate property value for loans and mortgages.

**Q4: What kind of data preprocessing was performed?**

Answer: Data preprocessing steps are handling missing values, encoding categorical variables, normalizing data, and feature selection to improve model accuracy.

**Q5: Which model performed the best in the evaluation?**

Answer: The XGBoost (Extreme Gradient Boosting) performed better than the other models in terms of accuracy and robustness.

**Q6: What performance metrics were used to evaluate the models?**

Answer: The models were evaluated using Root Mean Squared Error (RMSE), Mean Absolute Error (MAE) , Coefficient of Determination ($R^2$ Score) and MAPE (Mean Absolute Percentage Error) to measure prediction accuracy.

**Q7: What are the limitations of the current system?**

Answer: There are some limitations such as dependence on the quality and size of the dataset. The model is in inability to capture external market factors. Face challenges in generalizing to different geographic regions without retraining.

**Q8: How can the system be improved in future work?**

Answer: Future improvements would be integrating more diverse datasets using advanced models like Gradient Boosting or Neural Network. Considering external factors like economic indicators.

## Challenges

During this project, we encountered several challenges that required thoughtful solutions and teamwork. One of the main difficulties was data preprocessing. The dataset contained missing values, inconsistent formats, and outliers that needed to be handled carefully to avoid bias and maintain model accuracy. Choosing the right techniques for handling these issues, such as imputation, normalization, and encoding took considerable time and experimentation.

Another major challenge was featuring selection and engineering. Identifying which features had the most impact on house prices required deep analysis, including correlation studies and trial-and-error using different machine learning models. Some features were highly correlated or redundant, and selecting the optimal set while avoiding multicollinearity was critical for model performance.

We also faced technical limitations, especially when training complex models like XGBoost and ensemble methods, which required significant computational resources. Running multiple training and validation cycles for model tuning took time and occasionally caused memory or performance issues on limited hardware.

Additionally, evaluating model performance across different algorithms posed a challenge. While some models performed well on training data, they did not generalize well on test data, indicating overfitting. We had to carefully apply cross-validation techniques and fine-tune hyperparameters to strike a balance between bias and variance.

Lastly, integrating the machine learning model with the frontend interface was a non-trivial task. Ensuring that predictions were displayed accurately and interactively on a web-based platform involved debugging, API integration, and frontend validation.

Despite these challenges, we learned valuable problem-solving and teamwork skills, and were ultimately able to build a functional and accurate house price prediction system.

**Future Work**

To enhance the accuracy and reliability of the house price prediction model, future steps will focus on data enrichment, feature engineering, and model optimization. External data sources, such as geographic information, proximity to schools, business centers, and public transport and mortgage rates and housing demand trends will be integrated to provide a more comprehensive understanding of price determinants. Advanced feature engineering will be applied to create new attributes, such as total livable area and renovation history, while feature selection techniques will be refined to retain only the most impactful predictors.

Once the optimal model is finalized, deployment efforts will focus on creating a web-based application using Flask or FastAPI, making the prediction system accessible for real-world use. Furthermore, a model monitoring framework will be established to track performance over time, ensuring that the model adapts to market changes. These improvements will ensure scalability, accuracy, and long-term usability in real estate price estimation.

**Conclusion**

Predicting house prices accurately is a challenging but essential task, as it helps buyers, sellers, and real estate professionals make informed decisions. This project takes a machine learning approach to analyze key factors that influence property values, including home characteristics and market trends. Through careful data preprocessing, feature selection, and model optimization, we aim to build a reliable predictive system. However, some critical aspects, like location-based details and economic conditions, are missing from the dataset, which could improve accuracy if incorporated in future iterations.

Despite some challenges, such as handling missing data and optimizing model performance, significant progress has been made in building an effective prediction model. Moving forward, the focus will be on enhancing the dataset with additional features, fine-tuning model parameters, and deploying the system for practical use. By continuously refining the approach and integrating new data, this project has the potential to create a robust tool for estimating house prices, making real estate transactions more transparent and data driven.

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